



Cedar Initiative by NoxusDynamics LLP

PROJECT REPORT

on

Maize Leaf Disease Detection

TEAM NCD6

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1. Abstract

Maize is one of the most widely cultivated staple crops, and its yield is frequently threatened by leaf diseases such as Blight, Common Rust, and Gray Leaf Spot. Manual diagnosis of these diseases requires expert visual inspection and does not scale well for smallholder farmers. This project presents an automated image classification system that identifies the health condition of a maize leaf photograph as one of four categories: Blight, Common Rust, Gray Leaf Spot, or Healthy. The system uses transfer learning with a pretrained MobileNetV2 convolutional network implemented in PyTorch, and is deployed as an interactive Streamlit web application. The final model achieves a test accuracy of 91.10%, demonstrating that a lightweight, transfer-learning-based approach can deliver reliable, real-time disease diagnosis suitable for practical field use.

2. Problem Statement

Maize plants are commonly affected by leaf diseases that reduce crop health and yield. Identifying these diseases manually requires careful visual inspection and specialized agricultural knowledge, which is often difficult for non-experts and impractical at scale. The goal of this project is to build an image-classification model that can predict the condition of a maize leaf directly from a photograph, covering four classes: Blight, Common Rust, Gray Leaf Spot, and Healthy.

3. Solution

The proposed solution is an automated, image-based classifier that diagnoses maize leaf disease directly from a photograph. The approach uses transfer learning with a pretrained MobileNetV2 backbone, fine-tuned with a new 4-class classifier head, so the model can reuse rich visual features already learned on ImageNet rather than being trained from scratch. The trained model is packaged into an interactive Streamlit web application, allowing anyone to upload a photo and receive an instant, non-expert-friendly diagnosis along with a confidence score.

4. Methodology

4.1 Dataset

- Total images used: 4,188
- Blight: 1,146 images
- Common Rust: 1,306 images
- Gray Leaf Spot: 574 images
- Healthy: 1,162 images
- Split: 70% training (2,931 images), 15% validation (628 images), 15% testing (629 images)

4.2 Data Preprocessing & Augmentation

- Images resized to 224×224 pixels.
- Images converted to PyTorch tensors and normalized using ImageNet mean and standard deviation.
- Training augmentation: random horizontal flip, rotation, resized crop, and brightness/contrast changes.
- Dataset folders were audited and an incorrectly nested Blight folder was corrected.

- No tabular missing-value handling was required, since this is an image dataset.

4.3 Model Architecture — MobileNetV2

- Used a pretrained MobileNetV2 model from Torchvision.
- Transfer learning allows the model to reuse visual features already learned from ImageNet.
- Pretrained feature layers were frozen during training.
- The final classifier layer was replaced with a 4-output layer.
- Loss function: CrossEntropyLoss.
- Optimizer: Adam, with a learning rate of 0.001.

4.4 Training Process

- Training was performed for 10 epochs.
- The training set used augmentation; validation and test sets used only resize and normalization.
- Validation accuracy was checked after every epoch.
- The model with the highest validation accuracy was saved as best_maize_model.pth.
- Training was performed on CPU in a local VS Code environment.

5. Final Results

5.1 Accuracy

Best Validation Accuracy	Test Accuracy
91.24%	91.10%

- The model achieved similar validation and test accuracy, indicating reasonable generalization.
- The confusion matrix showed strong performance for Common Rust and Healthy leaves.
- Gray Leaf Spot was the most difficult class, partly because it had fewer training images.

5.2 Confusion Matrix

Rows represent the actual class; columns represent the predicted class.

Actual \ Predicted	Blight	Common Rust	Gray Leaf Spot	Healthy
Blight	158	2	16	0
Common Rust	9	187	0	1
Gray Leaf Spot	27	1	53	0
Healthy	0	0	0	175

- Healthy: 175 out of 175 images were correctly classified in this test split.

- Gray Leaf Spot was often confused with Blight (27 images).

6. Deployment: Streamlit User Interface

- A Streamlit web interface was created for easy testing.
- The user uploads an external maize leaf image.
- The app preprocesses the image exactly as required by the trained model.
- It displays the predicted class, confidence score, and probability for each class.
- The application is deployed online, so it can be tested without installing Python.

7. Tools & Technologies

- Python 3.14
- PyTorch & Torchvision
- MobileNetV2
- Pillow for image handling
- Matplotlib and Scikit-learn for evaluation
- Streamlit for the user interface
- VS Code for development
- Git & GitHub for version control and project sharing

8. Project Repository

The full source code, trained model, and Streamlit application for this project are available on GitHub:

<https://github.com/rifafat1ma/Maize-leaf-disease-detection>

A live, deployed version of the Streamlit application can also be tested here:

<https://maize-leaf-disease-detection-2yaanhpgbwghdgjy5mehh8.streamlit.app/>

9. Limitations & Future Improvements

- Gray Leaf Spot has fewer images than the other classes, creating class imbalance.
- Real-world images may differ from the training dataset in lighting, background, camera quality, and leaf orientation.
- Collect more field images, especially more Gray Leaf Spot samples.
- Fine-tune additional MobileNetV2 layers after the initial transfer-learning stage.
- Compare performance with EfficientNet or other lightweight architectures.
- Improve the interface and add disease information or recommended next steps.

10. Conclusion

This project demonstrates that a transfer-learning approach built on MobileNetV2 can accurately classify maize leaf images into four disease categories, achieving a test accuracy of 91.10% while remaining

lightweight enough to train on CPU hardware. Packaging the model as an interactive Streamlit application makes the solution accessible to non-expert users in the field. With additional data collection, particularly for the underrepresented Gray Leaf Spot class, and further architectural experimentation, the system has clear potential to be extended into a robust, field-ready crop health diagnostic tool.